

Remote Sensing Analysis of a Date Palm

Introduction

This study aims to assess the condition of a date palm farm using Sentinel-2 satellite imagery and QGIS software. Several spectral indices were calculated to evaluate vegetation health, chlorophyll content, water status, soil exposure, and temporal changes within the study area

Data Used

Satellite: Sentinel-2 Level-2A

Software: QGIS

Study Area: Palmira Farm, Siwa Desert Hinterland, Matrouh, Egypt

Spatial Resolution: 10–20 m

Methodology

The Sentinel-2 bands were processed in QGIS using Raster Calculator to generate several spectral indices

Results and Discussion

1 NDVI (Normalized Difference Vegetation Index)

Purpose

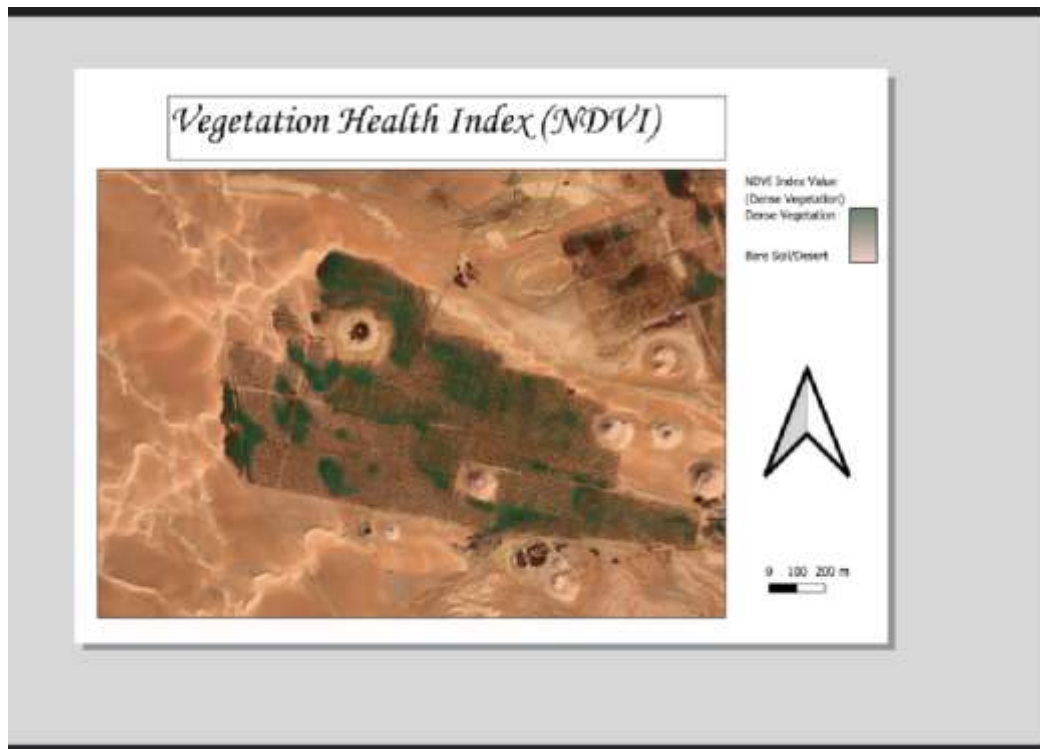
To evaluate the general health and vigor of date palm vegetation.

Formula

$$NDVI = \frac{(NIR - Red)}{(NIR + Red)}$$

Interpretation

- High values indicate healthy vegetation.
- Moderate values indicate stressed vegetation.
- Low values indicate sparse vegetation or bare soil.



2 gNDVI (Green Normalized Difference Vegetation Index)

Purpose

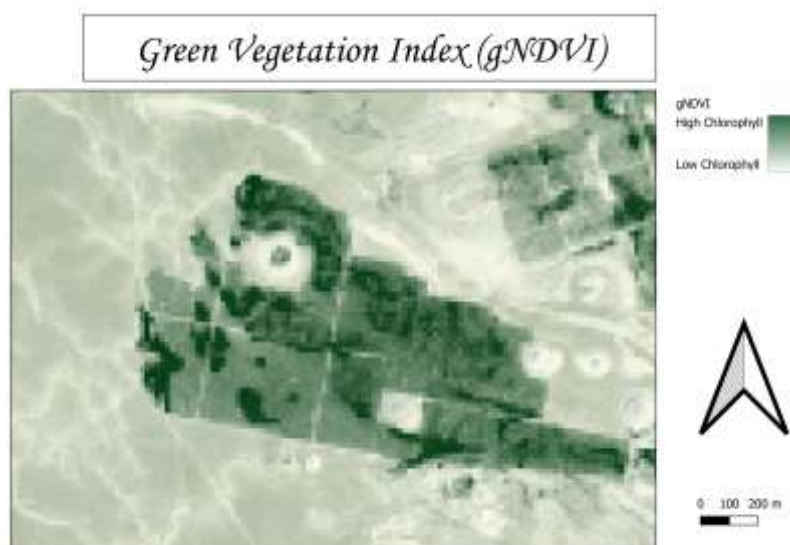
To assess chlorophyll concentration and plant vigor.

Formula

$$gNDVI = \frac{(NIR - Green)}{(NIR + Green)}$$

Interpretation

- High values indicate higher chlorophyll content.
- Low values may indicate nutrient deficiency or vegetation stress.



3 NDWI (Normalized Difference Water Index)

Purpose

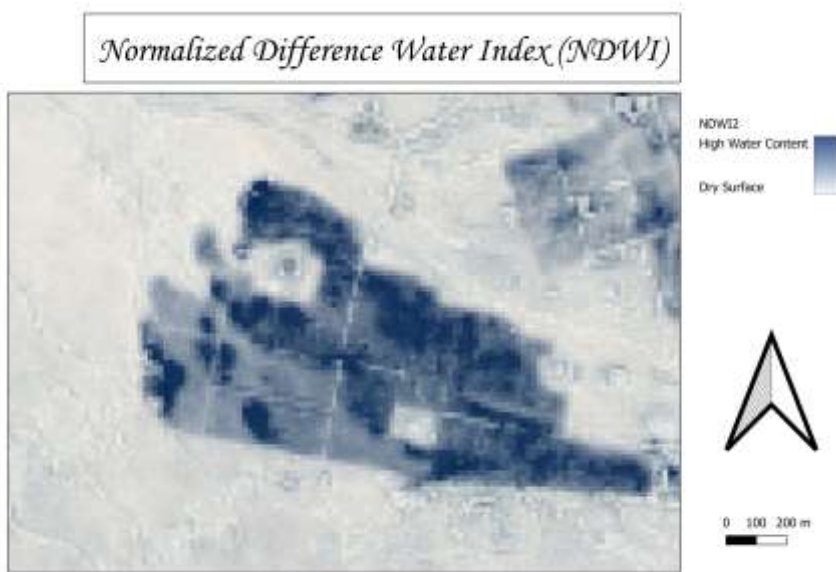
To evaluate moisture conditions and water content within vegetation.

Formula

$$NDWI = (NIR - SWIR) / (NIR + SWIR)$$

Interpretation

- Higher values indicate higher moisture content.
- Lower values indicate water stress and dry condition



4 SAVI (Soil Adjusted Vegetation Index)

Purpose

To assess vegetation health while minimizing the influence of exposed soil.

Formula

$$SAVI = (NIR - Red)(1 + L) / (NIR + Red + L)$$

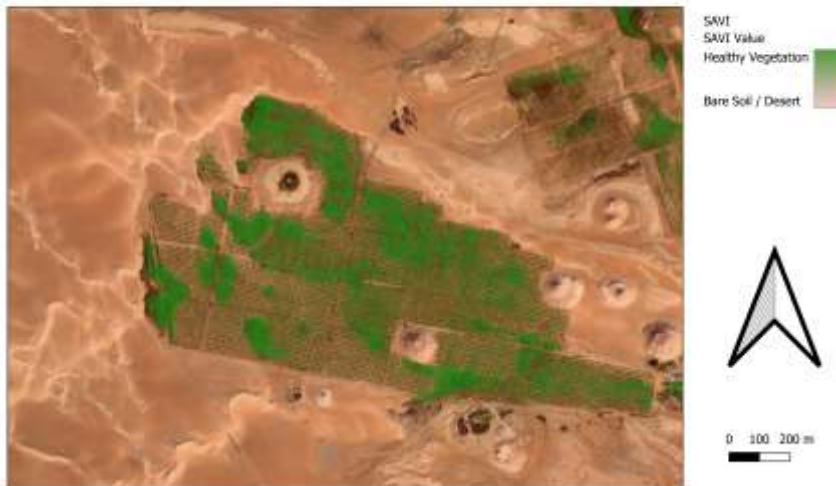
where:

$$L = 0.5$$

Interpretation

SAVI provides a more reliable vegetation assessment in arid and semi-arid environments such as the Siwa region.

Soil Adjusted Vegetation Index (SAVI)



5 BSI (Bare Soil Index)

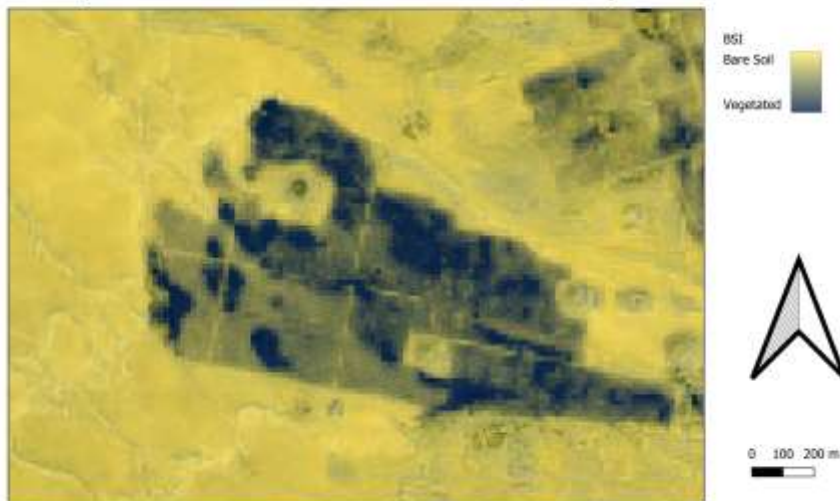
Purpose

To identify bare soil and exposed land surfaces.

Interpretation

- High values indicate bare soil and exposed ground.
- Low values indicate vegetated areas

Bare Soil Index (BSI)



6 Change Detection Analysis

Purpose

To identify temporal changes in vegetation conditions between two Sentinel-2 acquisition dates.

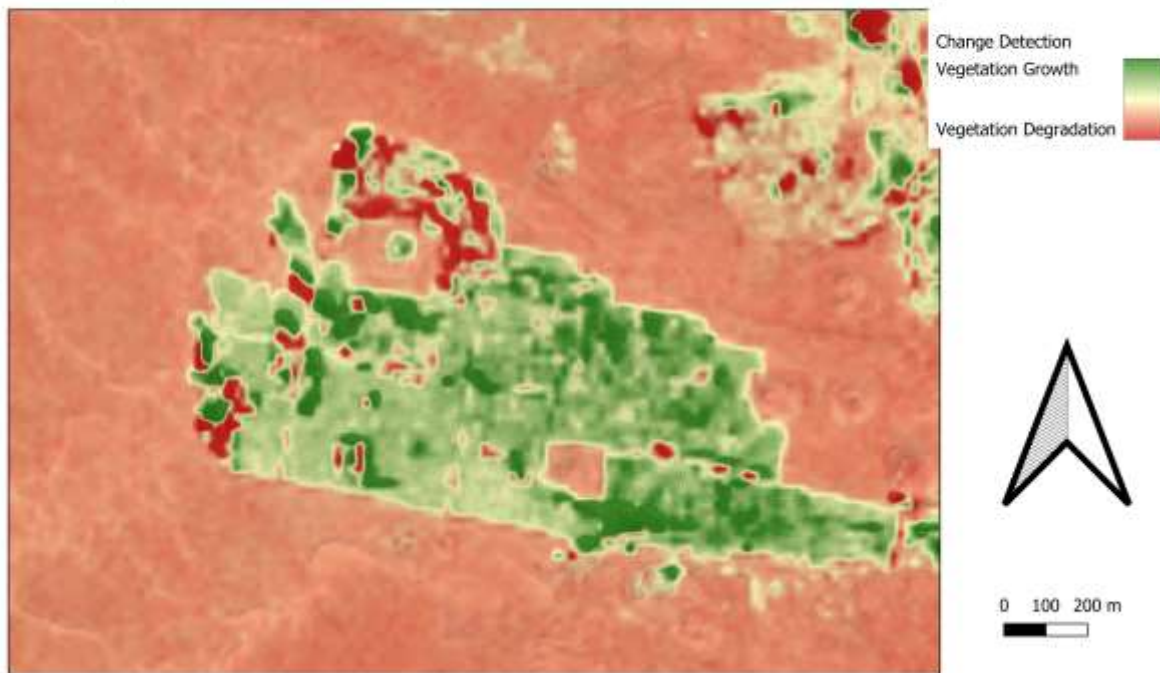
Method

Change detection was performed by subtracting the NDVI values of two different dates.

Interpretation

- Positive values indicate vegetation improvement.
- Negative values indicate vegetation degradation.
- Values close to zero indicate little or no change.

Agricultural Change Detection (2023 - 2026)



Conclusion

The spectral indices successfully characterized vegetation condition, chlorophyll content, moisture status, and soil exposure within the date palm farm. The change detection analysis provided valuable information regarding temporal variations in vegetation performance. These results demonstrate the usefulness of Sentinel-2 imagery and QGIS for agricultural monitoring and management in arid environments.
